THORACIC TRAUMA

Introduction & epidemiology

- **Second leading cause of trauma deaths.** 65–70% of chest injuries from RTAs. **~25% of trauma deaths due to chest injuries.** 50% of polytrauma deaths have significant chest injury.
- Immediate death = rupture of myocardium / thoracic aorta. **Early deaths (30 min–3h)** = tension pneumothorax, cardiac tamponade, airway obstruction, uncontrolled haemorrhage — often **preventable**.

Rib fractures

- Heal 3–6 weeks. Ribs 4–9 commonest.
- **Lower ribs (X–XII)** → **intra-abdominal injury (spleen, liver, kidney).**
- **1st + 2nd rib fracture + apical cap** → **great vessel injury** (high energy).
- Management: pain control, pulmonary function. **Elderly + multiple rib #** → **admit** (pneumonia risk).

Flail chest

- **≥3 consecutive ribs fractured in ≥2 places** → free segment → **paradoxical respiration** (moves IN on inspiration, opposite to normal).
- Associated **pulmonary contusion = the real problem**.
- Management: pain control, pulmonary toilet, O2, treat contusion. Severe → intubation + IPPV (internal pneumatic fixation).

Sternal fracture

- Anterior blunt / **seatbelt deceleration**. Low mortality.
- **Screen for myocardial contusion (ECG + serial cardiac enzymes), monitor 24–48h.**
- NO association with blunt aortic injury.

Subcutaneous emphysema

- Air in chest wall = marker of serious injury. Benign/self-limiting → **high-flow O2**. Massive → "gills" incisions.

Pulmonary contusion

- Alveolar oedema/haemorrhage; hypoxaemia ~50%; **X-ray findings within minutes, ARDS-like course**.
- Management: **restrict IV fluids**, pulmonary toilet, respiratory support. **NO prophylactic antibiotics.** Mortality 5–16%.

Pneumothorax

- **Simple**: hyperresonance, ↓breath sounds. Small/occult → observe; moderate/large → **chest tube (5th ICS, anterior axillary, safe triangle)**.
- **Open ("sucking chest wound")**: chest wall defect. → **three-way occlusive dressing** (prevents tension) → chest tube → definitive closure.
- **Tension (EMERGENCY)**: one-way valve → mediastinal shift. **Distended neck veins, hypotension, tracheal deviation to contralateral side, absent breath sounds, hyperresonance.**

- **CLINICAL diagnosis (before CXR). Immediate needle decompression: 2nd intercostal space, mid-clavicular line, 16G → then chest tube.**

Haemothorax

- Blood in pleura. **Lung = commonest source (self-limiting)**; intercostal/internal mammary/hilar = ongoing.
- ↓breath sounds + **dullness to percussion**. X-ray >200mL.
- **Chest tube. Thoracotomy if: initial >1500mL OR >200mL/h for 2–3h OR persistent transfusion need.**

Chest tube (ICD)

- **5th intercostal space, anterior axillary line — "safe triangle".** Blunt dissection over top of rib. Underwater seal = one-way valve; bottle below thorax.
- Indications: pneumothorax (tension/spontaneous/iatrogenic), haemothorax, haemopneumothorax, empyema, chylothorax, bronchopleural fistula.

Tracheobronchial injury

- MVC; **80% within 2.5cm of carina**; massive air leak, haemoptysis, sub-cut emphysema.
- **Persistent large pneumothorax + continuous air leak after chest tube = clue.**
- **Bronchoscopy = definitive diagnosis.** Intubation beyond injury; surgical repair. Mortality 10%.

Cardiovascular injuries

- **Myocardial contusion**: >55% of severe closed chest trauma. Monitor; **ECG + serial troponin** (normal at 4–6h excludes).
- **Penetrating cardiac**: **RV most common (43%; LV 34%)**; suspect in **"the BOX"** (clavicles–mid-clavicular lines–costal margins). 80% die at scene.
- **Cardiac tamponade**: **Beck's triad = muffled heart sounds + distended neck veins + hypotension**; narrow pulse pressure, pulsus paradoxus.
- **ECHO** (effusion + RV collapse); ECG electrical alternans. Fluids; **pericardiocentesis** (blunt/stable); **thoracotomy + repair** (penetrating).
- **Blunt aortic injury**: **most common vessel injured by blunt trauma. 80–90% at descending aorta distal to left subclavian (isthmus).**
- CXR: **widened mediastinum >8cm**, loss of aortic knob, apical cap, L effusion.
- **CT angiography = diagnosis.** Fix ABCDE first; **control BP (SBP 100–120, β-blocker/esmolol)**; surgery or endovascular repair. 85% survive if prompt.

Oesophageal perforation

- Penetrating; high morbidity, mediastinitis. **Gastric content in chest tube**, pneumomediastinum, effusion without chest injury.

Diaphragmatic tear

- Severe trauma + associated injuries → **mandates abdominal exploration**.

ED thoracotomy

- **Indications**: cardiac arrest in **PENETRATING** injury, massive haemothorax, penetrating + tamponade, large open wound, major vascular/tracheobronchial injury, oesophageal perforation. **CONTRAINDICATED in pulseless BLUNT trauma.**

- **5 moves**: 1. incise inferior pulmonary ligament · 2. open pericardium + staple/suture cardiac laceration · 3. open cardiac massage · 4. clamp pulmonary hilum / twist bleeding lung · 5. **clamp thoracic aorta**.

Thoracic trauma — high-yield

- **Tension pneumo = clinical dx → needle decompress (2nd ICS, mid-clavicular) → chest tube (5th ICS, safe triangle).**
- **Open pneumo = 3-way occlusive dressing → tube.**
- **Haemothorax thoracotomy: >1500mL or >200mL/h.**
- **Flail = ≥3 ribs ×2 places, paradoxical, treat contusion.**
- **Lower rib # → liver/spleen/kidney. 1st/2nd rib + cap → great vessel.**
- **Blunt aorta: distal to L subclavian; widened mediastinum; CT angio; β-blocker.**
- **Tamponade = Beck's triad; ECHO; pericardiocentesis. Penetrating = "the BOX", RV most common.**
- **Tracheobronchial: persistent air leak after tube → bronchoscopy.**
- **ED thoracotomy: penetrating arrest yes, blunt arrest no.**

PERITONEAL CAVITY & RETROPERITONEUM (IMAGING)

Core principle

- **CT = mainstay. US + MRI = no ionizing radiation.** Interpretation = anatomy + pathology principles.

Peritoneal cavity — ascites

- Small amounts detected US/CT/MRI; **nature NOT reliably differentiated**.
- US detects smallest amount. CT = low density.
- **Free ascites → dependent sites: pouch of Douglas, Morrison's pouch (hepatorenal), paracolic gutters.** Loculated → mimics abscess.

Peritoneal tumours

- Mostly metastases — **ovarian origin most important**.
- Imaging: (a) ascites only, (b) **peritoneal nodules**, (c) **omental cake**.

Intra-peritoneal abscess

- Follows **perforation**; site depends on perforation site; multiple common; ascites frequent.
- **US**: loculated fluid + irregular wall, internal echoes (septation/debris), air; DDx bowel (peristalsis).
- **CT**: fluid density + enhancing wall, **air in 50%** (bubble / streak / air-fluid level).
- **In-111/Tc-labelled leukocyte scan** if US/CT fail (e.g. anastomotic leak).

Retroperitoneum — anatomy

- **3 compartments** (anterior + posterior renal fascia). Organs: adrenal, aorta, retrocrural, IVC, psoas. **CT main**; plain film (mass/calcification/gas); MRI (multiplanar, vascular extension).

Retroperitoneal mass — DDx (7)

- 1. **Nodal** (lymphoma, germ cell, metastases)
- 2. Retroperitoneal fibrosis
- 3. **Neurogenic** (paraganglioma, schwannoma, ganglioneuroma)
- 4. **Sarcoma** (liposarcoma, fibrosarcoma)
- 5. **Haematoma** (trauma, ruptured AAA)
- 6. Infection (psoas/retroperitoneal abscess)
- 7. Vascular (aneurysm)

Lymphadenopathy

- Metastases & lymphoma look **identical** → **FDG-PET/CT** (esp lymphoma follow-up).
- Normal short axis: **para-aortic 10mm, retrocrural 6mm. >20mm = invariably malignant.**
- Lymphoma → enlarged nodes → confluent lobulated mass around aorta.

Adrenal imaging (CT best routine)

- **Non-functioning adenoma**: **<3cm hard to tell from mets; >3cm = mostly mets** (rarely carcinoma). Metastases common, **frequently bilateral, lung common primary**.
- **Cushing's adenoma**: always **>2cm, always seen, <10 HU**.
- **Aldosteronoma**: usually **<1cm — difficult to identify**.
- **Hyperplasia**: normal/mild enlargement.

- **Phaeochromocytoma**: very large, all detected; **10% rule (multiple, bilateral, extra-adrenal, malignant)**; **MIBG scan** localizes.

Retroperitoneal tumours

- Mainly liposarcoma/fibrosarcoma; nature usually undeterminable.
- **Liposarcoma recognizable = fat + strands of soft tissue density.**

Aortic aneurysm (AAA)

- **US = screening**; normal 2.5cm, **>3cm = aneurysm**. CT + US = true dimensions. Plain film lateral = calcification.
- CT: blood clot lining wall, calcification, IVC displacement.
- **>6cm = high rupture risk. Leaking aneurysm easily diagnosed on CT** (haemorrhage in retroperitoneum).

Imaging — high-yield

- **CT mainstay; US/MRI no radiation.**
- **Ascites: pouch of Douglas, Morrison's pouch, paracolic gutters.** Peritoneal mets → **omental cake** (ovarian).
- **Abscess: air in 50% on CT; leukocyte scan if CT/US fail.**
- **Node >20mm = malignant** (para-aortic 10mm); lymphoma ≈ mets on size → **FDG-PET**.
- **Cushing's adenoma >2cm (<10HU); aldosteronoma <1cm; phaeo 10% rule + MIBG.**
- **Liposarcoma = fat + soft tissue strands.**
- **AAA: US screen, >3cm aneurysm, >6cm rupture risk, CT for leak.**